



Order Matters

Quick facilitation notes & answer key

Goal

Students run two programs built from the SAME blocks in a different order and find that Bit lands on different numbers, then describe why and build their own order to a target. This is reading and altering code and describing how the change affects the outcome (C3.2), with a write-your-own step (C3.1).

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 so students can walk both codes and feel Bit stop at the end (5).

Run it (about 20 min)

1. I DO: walk Code 1 — 'start at 3... forward 3 would be 6, but the path ends at 5, so Bit stops on 5... back 1 to 4.'
2. WE DO: Exercise 1 — Code 1 lands on 4. Exercise 2 — predict Code 2, then run: back 1 to 2, forward 3 to 5.
3. YOU DO: Exercise 3 — same blocks landed on different numbers (4 vs 5); students explain why and write an order that reaches 5.
4. Connect: 'Computers run code in the exact order you give it. Change the order and you can change what happens.'

Answer key (modelled by the run-gate)

Ex 1 — Code 1: 3 → forward 3 stops at 5 → back 1 → 4.

Ex 2 — Code 2: 3 → back 1 → 2 → forward 3 → 5.

Ex 3 — (a) NO, not the same (4 vs 5). Why it works that way: in Code 1 the forward 3 happens at 3 and runs into the end of the path, so Bit stops at 5 and 'loses' a step; in Code 2 Bit steps back first, so forward 3 fits without hitting the end. (b) Any order that reaches 5, e.g. 'start at 3, back 1, forward 3' (the Code 2 order).

Watch for & differentiate

Common idea: students expect $3 + 3 - 1$ to always be 5; the edge of the path is what makes the order matter — model it on the floor path.

Simplify: do Code 1 and Code 2 side by side on the floor path so the different landings are felt, not just calculated.

Extend: ask for an order of the blocks that lands Bit on 4, and whether more than one order can reach the same number.

Success indicator: the child lands Code 1 on 4 and Code 2 on 5 and explains that the order changed the result.